

Contents

EPub-InkPlate - User's Guide - Version 3.0.0	1
1. Application startup	2
2. Interacting with the application	2
2.1 The Books List mode	3
2.2 The Book Reader mode	6
2.3 The Main Parameters Form	7
2.4 The Default Parameters Form	10
2.5 The Current book parameters form	10
3. Additional information	11
3.1 The books database	11
3.2 The Pages location computation	12
3.3 On the complexity of EPub page formatting	12
3.4 In case of out of memory situation	12
3.5 Images rendering	13
3.6 Moving the SD-Card from an Inkplate model to another	13
3.7 Internal fonts replacement	13
3.8 In case of a problem	14
3.9 Limitations	14

EPub-InkPlate - User's Guide - Version 3.0.0

The EPub-InkPlate is an EPub reader application built specifically for the InkPlate ESP32-based devices equipped with the Mechanical Buttons Extended Case. The extension can be found [here](#).

For the installation process, please consult the `INSTALL.pdf` document.

Here are the main characteristics of the application:

- TTF and OTF embedded fonts support
- Bitmap fonts support in a specific IBMF font format
- Normal, Bold, Italic, Bold+Italic face types
- Bitmap images dithering display (JPEG, PNG, GIF, SVG)
- EPub (V2, V3) book format subset
- UTF-8 characters
- Six mechanical keys to interact with the device
- Screen orientation (buttons located to the left, right, down positions from the screen)
- Linear and Matrix view of books directory
- Three size cover pictures user's selectable
- Up to 200 books are allowed in the directory
- Left, center, right, and justify text alignments
- Indentation
- Some basic parameters and options
- Limited CSS formatting
- WiFi-based documents download
- Battery state and power management (light, deep sleep, battery level display)
- Real-Time clock (All devices, but the Inkplate-6)

1. Application startup

When the device turns ON, the application executes the following tasks:

- Loads configuration information from the `config.txt` file located in the main SD-Card folder.
- Loads fonts definition as defined in the `fonts_list.xml` file located in the main SD-Card folder. Fonts must be located in the `fonts` folder on the SD-Card.
- Verifies the presence of books on the SD-Card, and updates its database if required. The books must be located in the `books` folder on the SD-Card, be in the EPub V2 or V3 format, and have a filename ending with the `.epub` extension in lowercase.
- Presents the list of books to the user. If a book was previously in progress, it will be opened at the last-read page.

2. Interacting with the application

Extended InkPlate devices have six mechanical buttons used to interact with the application. The screen orientation can be selected in the parameters form. Depending on the orientation, the function of each key changes as follows:

- When the keys are on the **Bottom** side of the screen, from left to right:
 - The **HOME** function
 - The **LEFT** function
 - The **UP** and **DOWN** function
 - The **RIGHT** function
 - The **SELECT** function
- When the keys are on the **Left** side of the screen, from top to bottom:
 - The **SELECT** function
 - The **UP** function
 - The **LEFT** and **RIGHT** function
 - The **DOWN** function
 - The **HOME** function
- When the keys are on the **Right** side of the screen, from top to bottom:
 - The **SELECT** function
 - The **UP** function
 - The **LEFT** and **RIGHT** function
 - The **DOWN** function
 - The **HOME** function

In the following text, the functions are called buttons.

The application has two main display modes:

- The Books List mode — Shows the list of books available on the SD-Card, displaying a small cover thumbnail, title, and author for each book.
- The Book Reader mode — Displays the book content one page at a time.

Each display mode also provides a set of functions the user can invoke. These are described in the sub-sections below.

2.1 The Books List mode

The list presents all books available to the user for reading. Two views are offered: a *linear view* and a *matrix view*:

- The Linear view will show books as a vertical list, showing the cover page on the left and the title/author on the right.
- The Matrix view will show covers arranged in a matrix with the title/author of the currently selected book at the top of the screen.

The application keeps track of the reading page location of the last 10 books opened by the user. A book will have its title prefixed with [Reading] to show this fact in the displayed list.

The books are presented in the following manner:

- Books being read are presented first in the list.
- The other books are then presented in alphabetical order by title.

The list may require several pages depending on the number of books present on the SD-Card and the size of the books' cover, selectable in the Main Parameters Form (see section 2.3 below).



Figure 1: The Books' List Linear View

King John *William Shakespeare*



Mon - 05/25 12:53

Figure 2: The Books' List Matrix View

Use the **UP** and the **DOWN** buttons to highlight the appropriate book that you want to read, then use the **SELECT** button to have the book loaded, presenting the first page of it.

The **LEFT** and **RIGHT** buttons can be used to move:

- one page at a time in the list for the linear view.
- one column at a time in the list for the matrix view.

The **HOME** button opens a list of options displayed at the top of the screen, each with an icon and a label beneath it. The options are as follows:



Figure 3: Books List options

- **Return to the e-books list** - This will simply get out of the options list, back to the list of books.
- **Return to the last e-book being read** - This will open the last book read by the user, to the last page shown on the screen.
- **Main parameters** - This will present the main parameters form, allowing the user to modify some elements of the application behavior. Its content is described below.
- **A** **Default e-book parameters** - This will present the default parameters form, allowing the user to modify some elements related to the book's reader. Those parameters constitute default values for books preparation and presentation. Its content is described below.
- **WiFi access to the e-books folder** - This starts the Wi-Fi connection and a web server, allowing you to manage the book list on the SD-Card from a web browser — uploading, downloading, and removing books. Once started, pressing any button stops the server and the Wi-Fi connection, and the device restarts. Note that while the web server is running, power-saving features (deep sleep and light sleep) are disabled.
- **Refresh the e-books list** - This refreshes the books database. The refresh happens automatically at startup and is rarely needed manually. Note that this action refreshes *all* books, which can be time-consuming — allow five to ten seconds per book.
- **Clear e-books' read history** - This erases all reading progress information (current position in each book and their priority placement at the top of the book list). The books themselves are not deleted.
- **Set Date/Time** - Opens a form to set the local date and time manually.
- **Retrieve Date/Time from Time Server** - This starts the Wi-Fi connection and retrieves the current time from an NTP server. The server address can be set in `config.txt`; the default is `pool.ntp.org`. Once the time is retrieved, press any button to restart the device.

- **i About the EPub-InkPlate application** - Shows a message box with the application version number and developer name.
- **⏻ Power OFF (Deep Sleep)** - Puts the device into Deep Sleep. Press any button to restart.

The **LEFT** and **RIGHT** buttons move the cursor between options. Press **SELECT** to execute the highlighted option. **HOME** closes the options list and returns to the book list (equivalent to selecting the first entry).

2.2 The Book Reader mode

The reader presents the book selected by the user one page at a time. Use the **UP** and **DOWN** buttons to go to the next or previous page. The **LEFT** and **RIGHT** buttons will go 10 pages at a time.

If the user presses the **UP** button when the first page of a book is presented, the reader will display the last page of the book. If the **DOWN** button is pressed when the last page of a book is presented, the reader will display the first page of the book.

As in the book list, the **HOME** button opens a list of options (the **SELECT** button does the same). The options are displayed at the top of the screen with an icon and label beneath each icon. The list is as follows:

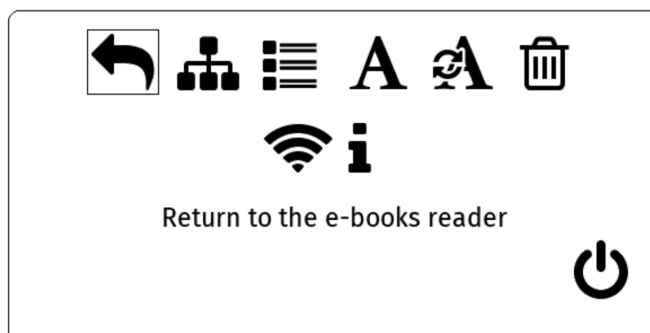


Figure 4: Book Reader options

- **↩ Return to the e-book reader** - Returns to the page being read in the current book.
- **📖 Table of Content** - If the book includes a table of contents, it will be shown here. Move the cursor to an entry and press **SELECT** to jump to that section. This option is only available if the EPub file contains a table of contents structure.
- **☰ E-Books List** - Exits the book reader and returns to the book list.
- **A Current e-book parameters** - Opens the parameters form for the current book, allowing you to select font and image display settings specific to this book. These settings are stored in a **.pars** file on the SD-Card. The available options are similar to those in the Default Parameters form, described below.
- **⌘ Revert e-book parameters to default values** - Resets all editable book formatting parameters to their default values.
- **🗑 Delete the current e-book** - Removes the current book and all its associated files from the device. A confirmation dialog is shown — press **SELECT** to confirm or any other button to cancel. After deletion, the book list is displayed.

- **WiFi access to the e-books folder** - This starts the Wi-Fi connection and a web server, allowing you to manage the book list on the SD-Card from a web browser — uploading, downloading, and removing books. Once started, pressing any button stops the server and the Wi-Fi connection, and the device restarts. Note that while the web server is running, power-saving features (deep sleep and light sleep) are disabled.
- **About the EPub-InkPlate application** - Shows a message box with the application version number and developer name.
- **Power OFF (Deep Sleep)** - Puts the device into Deep Sleep. Press any button to restart. If you were reading a book, it will be shown on screen when the device wakes.

The **UP/DOWN** or **LEFT/RIGHT** buttons move the cursor between options. Press **SELECT** to execute the highlighted option. **HOME** closes the options list and returns to the book list (equivalent to selecting the first entry).

2.3 The Main Parameters Form

As indicated in section 2.1, the Main Parameters form allows for the modification of some items available to the user that will change some application behavior. Each item is presented with a list of options selectable through the use of the keys.

Main parameters

Minutes Before Sleeping : 5 15 30

Books Directory View : LINEAR MATRIX

Books Cover Size : SMALL MEDIUM LARGE

Buttons Position (*): LEFT RIGHT BOTTOM

Pixel Resolution : 1Bit 3Bits

Show Battery Level : NONE
PERCENT
VOLTAGE
ICON

Show Title (*): YES NO

Right Bottom Corner : NONE DATE TIME HEAP

Battery Trim: 0.600

(*) Will trigger e-book pages location recalc.

Figure 5: The Main Parameters Form

The following items are displayed:

- **Minutes Before Sleeping** - Options: 5, 15, or 30 minutes. The idle timeout after which the device enters Deep Sleep, a state in which battery consumption is minimal. Once asleep, the device wakes on any button press.
- **Books Directory View** - Options: Linear or Matrix. This will select how the list of books will be presented to the user.
- **Books Cover Size** - Options: SMALL, MEDIUM, LARGE. Sets the size of the books cover that will be used to display the book list. They will be, respectively (Width x Height pixels):

70x90, 140x180, and 180x240 pixels.

- **Buttons Position** - Options: LEFT, RIGHT, BOTTOM. Sets the physical orientation of the device so the keys are located on the left, right, or bottom of the screen. Changing orientation between BOTTOM and LEFT/RIGHT (or vice versa) switches between landscape and portrait geometry, which affects how many characters fit on each page and may trigger a recalculation of page locations for all books.
- **Pixel Resolution** - Selects how many bits are used per pixel. 3 bits per pixel enables font anti-aliasing but requires a full screen refresh on every page turn. 1 bit per pixel enables faster partial screen updates but disables anti-aliasing, resulting in visibly jagged glyphs.
- **Show Battery Level** - Options: NONE, PERCENT, VOLTAGE, ICON. Displays the battery level at the bottom-left of the screen, updated each time the screen refreshes in the book list and book reader modes (not updated while option menus or parameter forms are displayed). PERCENT shows the charge percentage (2.5 V = 0%, 3.7 V and more = 100%). VOLTAGE shows the raw battery voltage. The icon is shown for all options except NONE. (A 3.7 volts rechargeable battery may have a value than can go up to around 4.2 volts)
- **Show Title** - When selected, display the book title at the top portion of pages.
- **Right Bottom Selection** - What to show at the bottom-right of the screen: nothing, the date/time, or the stack/heap size. When date/time is selected, it is shown as MM/DD - HH:MM (e.g., Mon - 01/24 22:44). When stack/heap size is selected, three numbers are shown left to right: unused stack space, largest available heap chunk, and total available heap memory. This is primarily useful for diagnosing memory issues. The total stack is 60 KB and the heap is approximately 4.3 MB.
- **Battery Trim** - This is a linear trim factor to adjust the proper display of the battery level. It must be set to a value between 0.0 and 2.0 exclusive (normally will be closer to 1.0). The battery level is being read by means of one of the ESP32's A2D (Analog to Digital) interface. This interface is known to have some limitation when reading analogic values. The resistors used to divide the voltage at the entry of that interface could also offset the value being read depending on their values. Here is the way to adjust the factor (if you are not familiar with electronics voltmeter, try to find somebody to help you):
 1. Set the **Show Battery Level** parameter to VOLTAGE;
 2. Return to the books directory view;
 3. Take note of the voltage that is shown at the bottom of the screen;
 4. Power Off the device;
 5. Open the device cover to get access to the zone where the battery is connected. Pay attention in the way you manipulate the device;
 6. With a DC voltmeter, read the voltage of the battery. There must be some circuit pads close to the battery connector that allow to read that voltage;
 7. You can now compute the proper trim factor using the following formula:
Voltage read on the voltmeter divided by the Voltage displayed on the screen

When the form appears, the currently selected option for each item is highlighted with a small rectangle. A larger thin-line rectangle — the *selecting box* — surrounds all options of the first item. (See Figure 5)

Main parameters

Minutes Before Sleeping : 15 30

Books Directory View : MATRIX

Books Cover Size : SMALL LARGE

Buttons Position (*):

Pixel Resolution : 1Bit

Show Battery Level : NONE
PERCENT

ICON

Show Title (*): NO

Right Bottom Corner : NONE HEAP

Battery Trim: 0.600

(*) Will trigger e-book pages location recalc.

Figure 6: Buttons Position is selected

To change an item, move the selecting box to it using **UP/DOWN** or **LEFT/RIGHT**, then press **SELECT**. The box changes to a **bold** rectangle around the options (see Figure 6). Use **UP/DOWN** or **LEFT/RIGHT** to cycle through choices, then press **SELECT** again to confirm. The box reverts to thin lines and advances to the next item.

Press **HOME** to exit the form. The newly selected options are saved and applied.

2.4 The Default Parameters Form

As indicated in section 2.1, the Default Parameters form (See Figure 7) allows for the modification of default values related to fonts and pictures usage. Each item is presented with a list of options selectable through the use of the keys.

Default e-books parameters

Default Font Size:

8101215

Line Height:

TIGHTMEDIUMLARGE

Use Fonts in E-books:

YESNO

Default Font:

DEJAVU SERIF
ROBOTO SANS
ROBOTO CONDENSED
OPEN SANS
IBARRA SERIF
CMU SERIF
CMU SANS
CMU TYPEWRITER

Show Images in E-books:

YESNO

Figure 7: The Default Parameters Form

The following items are displayed:

- **Default Font Size** - Options: 8, 10, 12, 15 points. Sets the character size for reflowable books (1 point = 1/72 inch).
- **Line Height** - Options: TIGHT, MEDIUM, LARGE. Sets the line height that will be used to display the text of the books.
- **Use Fonts in E-books** - Specifies whether fonts embedded in the book should be used to render pages.
- **Default Font** - Eight fonts are supplied with the application. Font names will have **CONDENSED**, **SERIF**, **SANS** (for sans-serif), or **TYPEWRITER** suffix to help distinguish the stroke of the font to use.
- **Show Images in E-books** - Controls whether images in books are rendered. Disabling images reduces memory usage and speeds up page rendering.

These are default values that apply to any book parameter that has not been customized for that specific book.

2.5 The Current book parameters form

As indicated in section 2.2, the current book parameters form allows for the selection of specific values related to fonts and pictures usage for the currently displayed book. Those parameters are specific to the currently displayed book and are kept in a `.pars` file on the SD-Card. Its content is similar to the Default parameters form of the books list as described in section 2.4.

The following items are displayed:

Current e-book parameters

Font Size: 8 10 **12** 15

Line Height Factor: TIGHT **MEDIUM** LARGE

Use fonts in book: YES **NO**

Font: DEJAVU SERIF
 ROBOTO SANS
 ROBOTO CONDENSED
 OPEN SANS
 IBARRA SERIF
CMU SERIF
 CMU SANS
 CMU TYPEWRITER

Show Images in book: **YES** NO

(Any item change will trigger book refresh)

Figure 8: The Current E-book Parameters Form

- **Font Size** - Options: 8, 10, 12, 15 points. Sets the character size for reflowable books (1 point = 1/72 inch).
- **Line Height** - Options: TIGHT, MEDIUM, LARGE. Sets the line height that will be used to display the text of the books.
- **Use Fonts in E-books** - Specifies whether fonts embedded in the book should be used to render pages.
- **Font** - Eight fonts are supplied with the application. Font names will have **CONDENSED**, **SERIF**, **SANS** (for sans-serif), or **TYPEWRITER** suffix to help distinguish the stroke of the font to use.
- **Show Images in E-books** - Controls whether images in the book are rendered. Disabling images reduces memory usage and speeds up page rendering.

When the form opens, it shows the values currently used to render the book's pages.

Parameters that the user has not explicitly set will show the value from the Default Parameters form. Once you change a parameter here, it is stored for this book. If you leave a parameter at its default, it will continue to track the value in the Default Parameters form — so updating the default will also update that book's presentation.

3. Additional information

3.1 The books database

The application maintains a small database of minimal metadata about each book (title, author, description, and a small cover image). This database is populated the first time the application detects a book on the SD-Card and is used to build the book list.

The only hard limit on the number of books is the capacity of the SD-Card. That said, a very large collection becomes difficult to browse — a few dozen books is comfortable, while a few hundred starts to feel unwieldy.

3.2 The Pages location computation

A book is displayed one page at a time. How much content fits on a page depends on the screen orientation (portrait or landscape), the fonts used, and the character size. The user-selectable parameters described in section 2 affect the number of pages and their positions within the EPub file.

When a book is opened, the application checks whether page locations need to be recalculated. If so, a background task handles this transparently, interfering minimally with reading and navigation. The page number shown at the bottom of the screen becomes available only once recalculation is complete. Locations are saved to a file so they do not need to be recomputed the next time the book is opened, provided the formatting parameters have not changed.

Page-location computation speed varies significantly with SD-Card speed. In tests with SanDisk Ultra cards (16 GB and 32 GB), each of the two supplied books took approximately 3 minutes to scan. With a slow older card (SanDisk 4 GB), the same scan took 8 minutes and 20 seconds.

3.3 On the complexity of EPub page formatting

The EPub standard exposes an enormous range of HTML/CSS formatting capabilities, and fitting a reasonable subset of that into a small processor is a real challenge.

I use a *good-enough* approach aimed at producing readable, enjoyable results. Some books with highly complex formatting will not render perfectly.

One practical remedy is the EPub converter in Calibre, a book management application available for Windows, macOS, and Linux. Using *Convert books* to convert an EPub to EPub simplifies its styling in a way that EPub-InkPlate handles more reliably.

The Calibre converter can also subset fonts to include only the glyphs needed by the book (in the *Convert books* tool: *Look & Feel* > *Fonts* > *Subset all embedded fonts*). One book I tested had four or five fonts totalling about 6 MB; after subsetting they came down to around 200 KB per font.

For images, select the *Generic e-ink* output profile under *Page setup* to resize images to match the InkPlate screen resolution (600×800 for InkPlate-6). Even at that size, a 600×800 image can take close to 500 KB.

Note that Calibre may not convert all images, and converted images will remain in RGB rather than grayscale, which increases load time. The script `adjust_size.sh` supplied with this release converts all embedded images to grayscale at a resolution of 800×600 pixels or smaller (you can modify it for 1200×825 for InkPlate-10). It requires **ImageMagick**, available on Linux, macOS, and Windows via **Cygwin**.

3.4 In case of out of memory situation

The memory available to prepare a book for display can be a limiting factor. InkPlate devices have approximately 4.5 MB of RAM, part of which is dedicated to the screen buffer.

For performance, fonts are loaded and kept in memory. A book that uses many fonts, or fonts that contain far more glyphs than the book actually needs, may exhaust available memory.

The following steps can help reduce memory usage:

- **Convert the book** - As described above, the Calibre converter can reduce both font and image size.
- **Use 1-bit pixels** - The frame buffer uses 240 KB at 3 bits/pixel and 60 KB at 1 bit/pixel (for an Inkplate-6). Select pixel resolution in Main Parameters.
- **Deactivate images** - In Main Parameters, you can disable image rendering.
- **Deactivate book fonts** - In Font Parameters, you can disable fonts embedded in the book.

If the application encounters a memory allocation failure, it will display a message explaining the cause and put the device into Deep Sleep. The message can guide you toward the appropriate step above.

3.5 Images rendering

Starting with version 1.3.0, the application uses a *stream-based* approach to render images, loading picture data incrementally from the EPub file to minimize memory usage.

JPEG, PNG, GIF, and SVG image types are supported, but only in their basic formats. Some books may contain images that cannot be rendered. The `adjust_size.sh` script included with the application can convert embedded images to a compatible format and resolution. See section 3.3 for details.

3.6 Moving the SD-Card from an Inkplate model to another

For each book, the application may generate three additional files in the `books/` folder of the SD-Card:

- Pages location offsets (files with extension `.locs`). They are tailored to the screen resolution and formatting parameters.
- Table of Content (files with extension `.toc`). They may also be tailored to the screen resolution and formatting parameters.
- The book's formatting parameters (files with extension `.pars`).

These files are automatically generated when they are not present (or when a formatting parameter will impact the page rendering) in the folder at the time the user opens a book to be read.

Different InkPlate models use eInk screens with different pixel resolutions. The application normally detects a resolution change and regenerates page locations when the user opens a book. If pages appear incorrectly after moving a card, delete all `.locs` and `.toc` files from the `books` folder on the SD-Card (plug the card into a computer to do this). The `.pars` files are compatible across all InkPlate models.

3.7 Internal fonts replacement

Starting with version 1.3.1, the application allows for the replacement of fonts that can be selected by the user through the configuration forms. To do so, a fonts configuration file named `fonts_list.xml` is used to define which font can be selected. This file must be present in the main SD-Card folder. It is loaded at boot time or after deep sleep to initialize the structure of the fonts.

Two groups of fonts are defined: SYSTEM fonts used to render application controls, and USER fonts that the user can select for displaying book content.

Each font in the **USER** group must specify normal, bold, italic, and bold-italic filenames. To avoid bad application behavior, the four combined font files for any single entry are limited to 300 KB. The **SYSTEM** group is tailored to the needs of EPub-InkPlate. Modifying it may affect how the application appears.

Font files must reside in the **fonts** folder on the SD-Card. The EPub-InkPlate distribution includes additional fonts not referenced in **fonts_list.xml** by default; you can add them by editing the XML file. Two font types are supported: True Type Fonts (**.ttf** files) and Open Type Fonts (**.otf** files). The previous versions of the application were also supporting the IBMF fonts, these were replaced with their equivalent Open Type Fonts. IBMF fonts are no longer supported.

Important: If you change fonts in the **USER** group, previously computed page locations may become invalid because glyph dimensions may differ. Delete all **.locs** files from the **books** folder; the application will recompute page locations automatically.

3.8 In case of a problem

If the application behaves unexpectedly, first check whether you are running the latest version using the **About the EPub-InkPlate application** menu entry. New releases are published at: <https://github.com/turgu1/EPub-InkPlate/releases>. Consult the Installation Guide for upgrade instructions.

If you are already on the latest release, connect the InkPlate to your computer with a USB cable and open a serial terminal. EPub-InkPlate sends diagnostic messages over the USB port while running, and any detected problem is likely to produce a message that identifies the cause.

On Linux and macOS, **minicom** works well for this. The device is typically **/dev/ttyUSB0**; use 115200 bps, 8N1.

On Windows, any serial terminal emulator will do. The device is typically **COM3**: with the same settings.

If you cannot resolve the issue on your own, open a report at: <https://github.com/turgu1/EPub-InkPlate/issues> — describe the unexpected behavior and attach any relevant diagnostic output.

3.9 Limitations

The Inkplate devices are based on ESP32-WROVER MPU. This is a very capable chip with a fair amount of processing power and memory. The following are the limitations imposed on the EPub-Inkplate application related to the capabilities available with the device.

- *Maximum number of books:* **200**. The application must keep some information about the books to quickly build and show the directory content.
- *Maximum single book size:* **25 Mbytes**.
- *Font formats:* **TTF, OTF**.
- *Maximum memory used for application internal fonts content:* **300 Kbytes**.
- *Maximum memory used for books' fonts content:* **800 Kbytes**. Fonts that are already loaded are kept for rendering. If the output is not appropriate, the user can disable the use of the fonts embedded with the book and use one of the fonts supplied with the application.

- *Maximum nested HTML tags in book content:* **50**. Testing the application, the author never had to deal with books having more than 15 nested tags. This limit is to track potential nested issues that would reset the device (stack overflow).
- *Image format types:* **subset of PNG, JPeg, GIF, and SVG**. The subsets are imposed by libraries used to interpret the image file content. In particular, JPeg pictures in progressive mode are not supported. They must be transformed to static mode using some tool (like Calibre) if you want them to be displayed.